Muscat 2040: Growth & Infrastructure Challenge

Technical Appendix & Calculation Logic

A) Data Inventory (Workbook Tabs Used)

All core calculations are derived from the provided dataset:

`MOH_health_units_data.xlsx ( a clearer form of the available data)`

Primary analytical tables used:

Workbook Tab	Purpose
Muscat Population (2002–2025)	Baseline population series used for projections
Muscat_omani_expart (2022–2024)	Population composition (Omani vs expatriate)
Average_Household_Size (2006–2018)	Household size trend used to estimate housing demand
muscat_moh_beds_manpower (2002–2023)	MOH hospital beds capacity baseline

Additional contextual tabs (interpretation only):

- `muscat_private_clinics` (2001–2020)
- `MOH Health Units Data` (1993–2010)
- `Electricity consumption` (2019–2023)
- `Oman population distribution` (2024)

These contextual datasets were **not required for numerical outputs**, but informed interpretation of infrastructure trends.

B) External Sources (Documentation Links)

External references are provided to document **definitions, benchmarks, and official statistical portals**.
All quantitative results in this submission are computed from the **provided workbook**.

OECD Hospital Bed Indicator

Definition and cross-country benchmark:

https://www.oecd.org/en/data/indicators/hospital-beds.html

OECD Health at a Glance 2023

International healthcare capacity context:

https://www.oecd.org/en/publications/health-at-a-glance-2023_7a7afb35-en/full-report/hospital-beds-and-occupancy_10add5df.html

Oman National Centre for Statistics and Information (NCSI)

Official population statistics portal:

https://data.ncsi.gov.om/?q=dataset%2Ftotal-population

https://data.gov.om/kglcksf/population-in-the-sultanate-of-oman

C) Assumptions Register

C1) Historical Population Growth

Population projections are based on the observed Muscat population series contained in the workbook.

The historical compound annual growth rate (CAGR) between **2002 and 2022** is calculated as:

$$CAGR = \left(\frac{Pop_{2022}}{Pop_{2002}}\right)^{\frac{1}{20}} - 1$$

Result

Average annual population growth:

≈ 5.27% per year

This long-run growth rate serves as the **baseline trend** for the projection model.

C2) Population Projection Assumptions

Parameter	Description	Value
Historical CAGR (2002–2022)	Long-run population growth rate calculated from Muscat population data	**5.27%**
Omani CAGR (2022–2024)	Recent growth rate for Omani population component	**3.25%**
Expat CAGR (2022–2024)	Recent growth rate for expatriate population component	**0.57%**
Blend weight	Weight applied to recent component trends when combining with long-run CAGR	**0.35**
Base Omani growth rate	Blended growth rate used in the base scenario	**4.56%**
Base expat growth rate	Blended growth rate used in the base scenario	**3.62%**
Anchor years	Observed population values used as fixed baseline before projections begin	**2022–2025**
Component split anchor	Year used to split population into Omani vs expatriate groups	**2025**
Expat population share	Share of expatriate population used in the anchor year	**59.17%**

Population growth is decomposed into **Omani and expatriate components** using the `Muscat_omani_expart` dataset.

$$Pop_t^{Omani} = Pop_t \times Share_t^{Omani}$$
$$Pop_t^{Expat} = Pop_t \times Share_t^{Expat}$$

Recent growth trends for both population groups are calculated and blended with the long-run historical CAGR to generate forward projections.

C3) Scenario Growth Parameters

Population projections are calculated using separate growth rates for Omani and expatriate populations under three scenarios.

Scenario	Omani Growth	Expat Growth
Low Growth	4.21%	2.82%
Base Growth	4.56%	3.62%
High Growth	4.91%	4.62%

These scenario parameters adjust **migration intensity and demographic growth** to create plausible population ranges through 2040.

C4) Healthcare Capacity Benchmark

Healthcare demand is estimated using the planning benchmark of:

3 hospital beds per 1,000 population

This benchmark is commonly used in **international health planning frameworks**, including OECD healthcare indicators.

For this analysis:

- Muscat hospital capacity is assumed to remain **constant at the latest observed value (2023)**.
- The dataset reports **1,608 MOH hospital beds** in Muscat.

Projected healthcare demand is compared against this capacity to estimate **system stress and potential future capacity gaps**.

C5) Housing Demand Assumptions

Housing demand is estimated by converting projected population into **households required**.

Because Muscat-specific household size data is not available in the dataset, the **Oman national household size series** is used as a proxy.

Parameter	Description	Value
Household size data source	Oman national household size series from workbook	2006–2018
Household size estimation method	Linear trend extrapolation	Linear regression
Intercept (a)	Regression intercept used in projection	75.121
Slope (b)	Annual change in household size	−0.0337 per year
Household size bounds	Minimum and maximum limits applied to avoid unrealistic projections	3.5 – 9.0 persons

Household demand is calculated as:

$$Households_t = \frac{Population_t}{Household\ Size_t}$$

Because the dataset does not include **Muscat housing stock information**, the analysis estimates **housing demand (households required)** rather than a full housing supply gap.

Data Limitation Note

Average household size in the workbook is **Oman-wide rather than Muscat-specific**. Therefore, the national series is used as a **proxy** when estimating housing demand in Muscat.

D) Step-by-Step Calculations

D1) Population Projection (2023–2040)

Step 1 — Baseline

Observed population is taken from the **Muscat Population** sheet.

Step 2 — Historical Growth

Compute CAGR using the 2002–2022 series.

$$CAGR = \left(\frac{Pop_{2022}}{Pop_{2002}}\right)^{\frac{1}{20}} - 1$$

Step 3 — Composition Split

Using Omani/expatriate shares:

$$Pop_t^{Omani} = Pop_t \times Share_t^{Omani}$$
$$Pop_t^{Expat} = Pop_t \times Share_t^{Expat}$$

Step 4 — Recent Component Growth

Calculate **2022–2024 CAGR** separately for:

- Omani population
- Expat population

Step 5 — Scenario Growth Rates

Growth rates are generated by blending:

- Long-run CAGR
- Recent component growth

Then adjusted for **Low / Base / High** scenario ranges.

Step 6 — Anchoring Observed Years

Observed totals are kept fixed for:

2022–2025

Projections begin in:

2026

Step 7 — Forward Projection

For each year:

$$Pop_{t+1}^{Omani} = Pop_t^{Omani}(1 + g_O)$$
$$Pop_{t+1}^{Expat} = Pop_t^{Expat}(1 + g_E)$$

Total population:

$$Pop_t^{Total} = Pop_t^{Omani} + Pop_t^{Expat}$$

D2) Healthcare Demand vs Capacity

Bed Demand

For each scenario and year:

$$BedsDemand_t = Pop_t \times \frac{BedsPer1000}{1000}$$

Where:

BedsPer1000 = 3

Capacity

Healthcare capacity is fixed at:

MOH Muscat beds (2023)

from:

`muscat_moh_beds_manpower`

Gap and System Stress

Bed gap:

$$Gap_t = BedsDemand_t - BedsCapacity$$

Stress index:

$$Stress_t = \frac{BedsDemand_t}{BedsCapacity}$$

Breakpoint Year

Defined as the **first year where**

$$Gap_t > 0$$

This indicates the year when healthcare capacity becomes insufficient.

Output file:

`outputs/tables/healthcare_beds_demand_gap.csv`

D3) Housing Demand (Households Required)

Household Size Projection

Household size is estimated using a **linear fit** on:

`Average_Household_Size`

Household Demand

$$Households_t = \frac{Pop_t}{HHSize_t}$$

This estimates the number of **housing units required to support population growth**.

Output file:

`outputs/tables/housing_households_needed.csv`

E) Reproducibility Instructions

Install dependencies

```
python -m pip install -r requirements.txt
```

Generate analysis tables

```
python analysis\build_outputs.py
```

Generate figures

```
python -m analysis.make_figures
```

Launch interactive model

```
streamlit run streamlit_app/app.py
```

F) Files Produced (Submission Bundle)

Tables

```
outputs/tables/*.csv
```

Includes:

- population projections
- healthcare demand gaps
- housing demand estimates
- assumptions register

Figures

```
outputs/figures/*.png
```

Charts include:

- population projection scenarios
- hospital bed demand vs capacity
- housing demand growth

Interactive Model

```
streamlit_app/app.py
```

Allows reviewers to:

- adjust growth assumptions
- explore scenario outcomes
- visualize infrastructure stress points

Narrative Documents

```
outputs/Muscat_2040_Executive_Summary.md
outputs/Muscat_2040_Technical_Appendix.md
```

G) Model Limitations

While the model provides transparent estimates of infrastructure demand, several data limitations should be noted:

Housing stock data
The provided dataset does not include Muscat's current housing inventory.
As a result, this analysis estimates **housing demand (households required)** rather than a full supply gap.

Household size proxy
Muscat-specific household size data was not available in the workbook.
The model therefore uses the **Oman national household size series** as a proxy.

Healthcare capacity scope
Healthcare capacity estimates include **MOH hospital beds only**, based on the `muscat_moh_beds_manpower` dataset.
Private hospital capacity and future planned facilities were not included due to limited publicly available data.